

A TERMINAL BATTLE GAME

tetrish.dev

PLAYER MANUAL



INSPIRED BY TETRIS BATTLE SAIDEN

DORESTACK CHALLENGE 2026 / MAO MINI TEAM

QUICK START

1

INSTALL KITTY + CLONE THE REPO

Open Kitty on your Linux machine / VM or macOS. Install it First if needed, then clone the repository.

```
sudo apt install kitty
```

```
brew install kitty
```

```
sudo pacman -S kitty
```

```
git clone https://github.com/ziqili/MacMini_tetrISH.git
```

2

ENTER THE REPO + OPEN THE CLIENT

Go to the root of the cloned repository, then build and launch the player client.

```
cd MacMini_tetrISH
```

```
make play
```

3

SIGN IN OR PLAY OFFLINE

Create an account, sign in, or choose Play Offline for a Fast Solo warm-up.

4

CHOOSE YOUR MATCH

Warm up in Solo, duel in Double, or enter Battle Royale. Pick a Fighter before the countdown.

5

CLEAR, CHARGE, DASH

Every two cleared lines bank one charge. Press 1-4 when a power is ready.

6

CLAIM THE FREE EXTRAS

If you want them, open Marketplace to claim the Design AI theme and Mirurun for Free, then equip them in Settings.



2 LINES = 1 CHARGE

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SEVEN PIECES // FOUR FIGHTERS // ONE STACK



I



O



T



S



Z



J



L

ROSE / ACTION

PLUM / SYSTEM

COLOUR / GAME

WELCOME TO tetriSH.dev

A CONSOLE GAME HIDING IN A TERMINAL

tetriSH.dev began with a slightly unreasonable idea: what if a whole multiplayer Tetris game lived inside a terminal, but still felt like something you would find in a game case? This booklet is your map. Pick a fighter, build clean, and make the right power hurt at exactly the right time.

THE FEEL

Fast Tetris, loud character powers, pixel art and real terminal constraints.

THE LOOP

Place pieces, clear lines, bank charge, then spend it on a level 1-4 power.

THE RULE

Your client asks. The server decides. Only confirmed state is drawn.

YOUR FIRST GOOD DECISION

Do not fire a power the instant it lights up. Watch the next pieces, the height of your stack, and the rival you are targeting. A level 1 escape can save a match; a level 4 cast can end one. Charge is a choice, not just a meter.

BUILD LOW // READ AHEAD // SAVE A CAST



START THE STACK

1 SERVER + DAEMONS

Build the stack and start `tetrish`, `tetrislogd` and `tetrisd` from the repository root.

```
make run
```

2 CHECK THE STACK

Confirm that the logger and authoritative game server are alive before connecting.

```
tetrisctl status
```

3 BUILD + OPEN CLIENT

In a second terminal, build `tetrisu` and launch the player-facing game entry point.

```
make play
```

4 STOP CLEANLY

When the session is over, ask the control plane to stop the managed daemons cleanly.

```
tetrisctl stop
```

WHAT `make run` HANDLES

```
tetrish > .tetrishrc > tetrislogd > tetrisd
```

```
tetrisctl status // both daemons report ready
```

The logger starts first, so even the server's first breath can be recorded.

PLATFORM NOTE

Run `tetrisd` on Linux or WSL. A macOS build can run `tetrisu` and connect to that server.

CONTROLS + FIELD

GAME KEYS

LEFT / RIGHT

MOVE

held keys repeat

DOWN

SOFT DROP

Faster Fall

UP / X

ROTATE CW

clockwise

Z

ROTATE CCW

counter-clockwise

SPACE

HARD DROP

lock Fast

O

HOLD

swap the piece

1-4

DAST POWER

cost 2 / 4 / 6 / 8

P

PAUSE

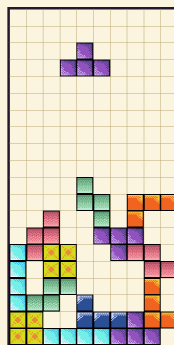
Solo only

Q / ESC

LEAVE

confirmation in Solo

10 COLUMNS



20 ROWS

NEXT



HOLD



ROYALE TARGETING

W

A

S

D

KOs / RANDOM / ATTACKERS / BADGES

CHARGE LADDER



LV.1 = 2 // LV.2 = 4 // LV.3 = 6 // LV.4 = 8

SOLO MODE

Solo is where you learn how tetrISH.dev breathes: the gravity curve, the hold slot, the charge rhythm and the timing of a Fighter kit. Online Solo still follows the same rule as every match - the server owns the board.

THE SOLO HUD

SCORE

018480

LINES

18

LEVEL

04

CHARGE

HOLD

I

NEXT

↑ / L / O

FIGHTER

MIRURUN

PRESS 1-4 TO CAST

8 LINES = +10CHARGE

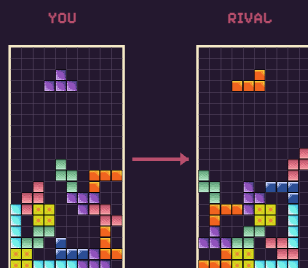
FIRST-MATCH ROUTINE

- 1 Keep the stack flat enough for every piece.
- 2 Hold the awkward piece; do not panic-place it.
- 3 Watch charge before choosing a risky setup.
- 4 P pauses Solo. R restarts after game over.

DOUBLE + BATTLE ROYALE

DOUBLE

A clean duel. Create or join a room, look a fighter, and let the owner start the countdown. Your one rival is the natural target for every offensive power.



CREATE / JOIN > LOOK FIGHTER > OWNER STARTS > LAST BOARD STANDING

There is no pause in a live match. Leaving the local board frozen while a rival keeps moving would not be a pause - it would be a loss.

BATTLE ROYALE

The same board, now surrounded by rivals. Multi-line clears create pressure, and the targeting diamond decides who receives it. Stay low, read the room, and save a defensive cast for the ugly moment.

YOUR FIELD



TARGET DIAMOND



CLEAR N >= 2 // SEND N - 1

PROFILE + THEMES

PLAYER PROFILE

PLAYER_0042

BEST SCORE 86,400

RANK #12

WALLET 23

FIGHTER PRINCESS

THE ECONOMY

Your best single-game score decides rank. Lifetime score feeds the wallet at 1 wallet point per 100 scored, then purchases are deducted.

PLAY

EARN

BUY

EQUIP

Characters cost 10. Themes are cosmetic only.

THEME CABINET

ARTWORK + MUSIC + NICKNAMES // POWERS STAY THE SAME

DEFAULT

FREE



DESIGN + AI

FREE



SNOWMAN

5 WALLET PTS



CLAUDE-ING

7 WALLET PTS



HAALAND

8 WALLET PTS



AL-MERQAEDES

12 WALLET PTS



NUCLEAR GANDHI

12 WALLET PTS



EQUIP + SAVE

BUY: MARKETPLACE // EQUIP: SETTINGS // SAVED

TERMINAL TIER LIST

tetrisH.dev probes the terminal and chooses the safest renderer. Better tiers move authored pixel planes; every tier can still play.

S MOVABLE	KITTY GHOSTTY
A STATIONARY	WEBTERM ITERM2 SIXEL LINUX FB
B CELL SAFE	ANY NOTOURES TERMINAL WITHOUT PIXEL SUPPORT

FORCE A TIER FOR TESTING

...=pixel

...=stationary

...=cell

VARIABLE: TETRISU_RENDERER

CLIENT: macOS / LINUX // SERVER: LINUX OR WSL

THE GAIDEN CONNECTION

ORIGINAL // 1993

Tetris Battle Gaiden gives both players one shared three-piece preview. Crystal pieces fill a meter, and each Fighter spends that meter on four character powers.

SHARED PREVIEW // CRYSTAL POWER

tetrish.dev // ADAPTATION

tetrish.dev keeps the four fighters and their power personalities, but changes the fuel. Clear lines to bank charge; crystal blocks are not part of its game logic.

2 LINES = +1 // BOOST 2 / 4 / 6 / 8

TWO POWER FLOWS



LAND CRYSTAL > FILL > OAST



CLEAR > BANK > PRESS 1-4

WHAT CARRIED OVER

ROSTER

FOUR FIGHTERS

KITS

FOUR POWERS EACH

MIND GAME

READ / SAVE / OAST

NEXT // MEET THE FOUR FIGHTERS





FIGHTER // 01

HALLOWEEN

FRY // DARK // VAMPIRE // BOMB

Pick Halloween when you want the whole match to misbehave. Burn your own board clean, hide a rival's Field, steal charge or scatter their stack.

ALL FOUR POWERS // PLAYABLE IN THE FULL GAME

LV.1 // 2 CHARGE

FRY

SELF
NEXT LOOK

Add three hot rows to your own Field.
Lock the next piece and they burn away.



LV.2 // 4 CHARGE

DARK

RIVAL
BLACKOUT

Pull a blackout over the rival Field,
leaving only a small window beneath
the active piece.



LV.3 // 6 CHARGE

VAMPIRE

RIVAL
CHARGE STEAL

Take the rival's stored charge for
yourself. Their partial line progress
stays put.



WHOLE CHARGE

LV.4 // 8 CHARGE

BOMB

RIVAL
STACK DAMAGE

Knock a scattered set of cells out
of the rival stack.





FIGHTER // 02

MIRURUN

MIRURUN // INVERSION // PENTARIS // SIRTET

Mirurun looks harmless. The kit is anything but: one clean escape button, followed by three wonderfully mean ways to scramble a rival.

ALL FOUR POWERS // PLAYABLE IN THE FULL GAME

LV.1 // 2 CHARGE

MIRURUN

SELF
BOTTOM OUT

Lift away your bottom four rows. Nothing is sent to a rival.



LV.2 // 4 CHARGE

INVERSION

RIVAL
3 PIECES

Swap left and right for the rival's next three locked pieces.

LEFT RIGHT



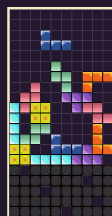
SWAPPED x3

LV.3 // 6 CHARGE

PENTARIS

RIVAL
5 GARBAGE

Send five garbage rows to the rival field, each with a playable hole.



LV.4 // 8 CHARGE

SIRTET

RIVAL
ROW INVERT

Flip occupied rows on the rival board: spaces fill while blocks disappear.





FIGHTER // 03

PRINCESS

SOL // MIRROR // PARALYSIS // COPY

Princess rewards precision: carve a lane through your own stack, then answer rival powers with clever counters.

ALL FOUR POWERS // PLAYABLE IN THE FULL GAME

LV.1 // 2 CHARGE

SOL

SELF
AIM A COLUMN

Aim at one column. Sol clears it and the two beside it from your field.



LV.2 // 4 CHARGE

MIRROR

RIVAL
NEXT POWER

Catch the next power aimed at you and steal it for yourself.



STEAL NEXT POWER

LV.3 // 6 CHARGE

PARALYSIS

RIVAL
3 PIECES

Stop the rival from rotating their next three pieces.



NO ROTATE x3

LV.4 // 8 CHARGE

COPY

RIVAL
FIELD COPY

Replace your field with a copy of the rival field.





FIGHTER // 04

WOLF-MAN

OUT // NUE // PALS // THWACK

Wolf-man is the survivor: shave danger off the top, slow a rival, soften garbage and set up cascade clears.

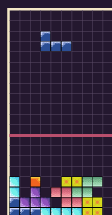
ALL FOUR POWERS // PLAYABLE IN THE FULL GAME

LV.1 // 2 CHARGE

CUT

SELF
TOP OUT

Slice the top four rows off your own field.



LV.2 // 4 CHARGE

NUE

RIVAL
4 PIECES

Block the rival's soft and hard drops for their next four pieces.



LV.3 // 6 CHARGE

PALS

SELF
GARBAGE GUARD

For a short time, ordinary rival garbage lowers your stack. Power-made lines are excluded.



LV.4 // 8 CHARGE

THWACK

SELF
4 PIECES

For your next four pieces, a clear lets loose blocks fall again and open chain clears.



INSIDE THE STACK

YOU PRESS. THE SERVER DECIDES.



STATE RETURNS // ONLY CONFIRMED DATA IS DRAWN

The active branch uses an epoll reactor for established connections, plus a bounded handshake pool so slow handshakes cannot own the server.

AUTHORITY RULES

- > Client input never edits the final board directly.
- > The server checks identity, room, target and charge.
- > Game logic is pure: I/O stays in the daemon.
- > A refused cast spends no charge.

CONCURRENCY + IPO

- > Room state is protected while it changes.
- > Locks are released before slow network writes.
- > Log records enter a non-blocking ring buffer.
- > tetrislogd writes them in another process.

PROGRAMS + FILES

FIVE PROGRAMS

tetris.h

custom shell + startup

tetrisu

terminal game client

tetrisd

authoritative game server

tetrislogd

separate log writer

tetrisctl

daemon lifecycle + admin

EIGHT STATIC LIBRARIES

libtetrisbrain

pieces, scoring, charge, effects

libtetrisroom

lobby, seats, room lifecycle

libmaominldb

accounts, wallet, inventory, WAL

libtetrissh

certificate auth + AES Frames

libhttp

protocol parser + serialiser

libstatusbody

STATE/profile body codecs

liboorelpo

rings, Unix sockets, message queues

libooredaemon

detach, pidfiles, readiness

RCPO MAP

src/

programs + daemons

lib/

self-contained static libraries

docs/

design, tests + bugs

assets/

themes, sprites + sound

tmp/

runtime state

certs/

development identity

HELP + WHAT COMES NEXT

QUICK FIXES

NO CONNECTION

Check the daemons, port and certificate files before blaming the client.

```
tetrisc21status // make certs
```

ART LOOKS WRONG

Resize the terminal or force the renderer tier that matches its graphics support.

```
TETRISU.RENDERER=cellstationary|pixel
```

SERVER ON macOS

The current server uses epoll and POSIX message queues. Run it on Linux or WSL; macOS can be the client.

NO AUDIO

SDL audio is optional. tetrISH.dev degrades to silence; gameplay and networking continue.

POST-v1 IDEAS // NOT CURRENT FEATURES

macOS SERVER

kqueue + portable IPQ

RECONNECT

resume a dropped match

SPECTATE

watch rooms and replays

TERMINAL LAB

automated backend matrix

TOURNAMENTS

brackets + room discovery

CONTROL PLANE

richer local admin commands

TURN THE PAGE // THE INSIDE-BACK CHEAT SHEET IS YOUR LAST SAVE POINT

KEEP THIS OPEN

GAME

ARROWS

move + soft drop

X / Z

rotate CW / CCW

SPACE

hard drop

O

hold

1 - 4

Fighter powers

P

pause Solo

Q

leave match

ROYALE

W

target KOs

A

target randoms

S

target attackers

D

target badges

+ / -

audio volume

CLICK

select a power

NO P

live matches do not pause

STACK

server

make run

client

make play

start

tetrissetl start

status

tetrissetl status

stop

tetrissetl stop

restart

tetrissetl restart

certs

make certs

FIGHTER QUICK PICK

PORTRAIT // FIGHTER



HALLOWEEN

CONTROL



MIRURUN

CHAOS



PRINCESS

PRECISION



WOLF-MAN

DEFENSE

LEVEL 1+2 // LEVEL 3+4

1 FRY: CLEAR SELF // 2 DARK: BLACKOUT

3 VAMPIRE: STEAL CHARGE // 4 BOMB: BREAK STACK

1 MIRURUN: OUT BOTTOM // 2 INVERSION: REVERSE

3 PENTARIS: +3 GARBAGE // 4 SIRET: FLIP ROWS

1 SOL: CLEAR 3 COOLS // 2 MIRROR: STEAL POWER

3 PARALYSIS: LOOK ROTATE // 4 COPY: RIVAL FIELD

1 OUT: REMOVE TOP 4 // 2 NUG: BLOOD DROPS

3 PALS: GARBAGE GUARD // 4 THWACK: CASCADES

OUTSIDE BACK // PLAYER MANUAL

THANKS FOR PLAYING

Built in 0 From the shell prompt to the Falling piece: one small terminal
game, one very large stack, the corestack!



SOURCE REPOSITORY

github.com/ziqiiii/MacMini_tetrish

MAC MINI TEAM

SANJAN KRISHNA SARAT // THONG BI QI

SUTD \$0.003 x \$0.00\$ CORESTACK CHALLENGE // 2026

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